ADEPT · TEAM PECEL LELE

SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

1. Team & Project Overview

Field	Response
Team Name	Pecel Lele
Project / Product Name	Adept
Final Live Deployment URL	https://adept-my.replit.app
Codebase Repository URL	https://github.com/pecel-lelele/adept
Name (Leader)	David Kurniawan
Email & phone (Leader)	*To be completed by team leader*

2. Project Description [Maximum 250 words]

Adept is a human-led, sourced reskilling-planning tool. A learner confirms skills from a form or synthetic demonstration CV, explores dated AI-research signals, compares catalogue-backed roles and gaps, then chooses an ordered course plan within time and budget. Moderate and Fast are assumption-based scenarios, not calibrated employment forecasts. Research attention and a US labour baseline are proxies; Malaysian job listings and national labour statistics are displayed with their own scope and dates, not substituted for a Malaysian role-demand forecast.

Employers or career centres can invite members, view consent-gated aggregate gaps, propose programmes and review progress. Members decide whether to share skills and accept a programme; private plans remain separate. Role/skill links to UN SDG targets are curated explanations, not verified SDG outcomes. Adi answers how-to questions from fixed, reviewed answer blocks; an optional language model may only select block IDs. Recommendations carry source, snapshot, scenario and estimate metadata. The application does not hire, fire or certify anybody. The competition demo uses public reference records and synthetic people only. Adept's intended first commercial workflow is an L&D manager choosing which training to fund for a small operations cohort; demand and willingness to pay will be validated in a paid pilot.

3. Upgraded Product & Technical Execution

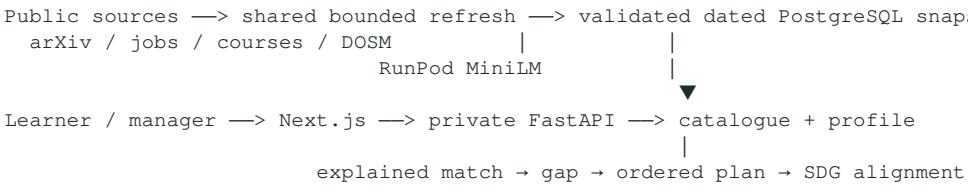
3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

A single public Next.js 16 server presents learner, employee and manager views and proxies requests to a private FastAPI/Python 3.11 process. Signed accounts, profiles, memberships, catalogue references, snapshots, plans and evidence live in Replit's managed PostgreSQL; server-side membership, manager and consent checks guard organisation data. The local/CI database is PostgreSQL 16 with pgvector. Production

startup checks readiness but does **not** migrate or seed.

Daily GitHub Actions supplies a clock and authenticated trigger; the deployed application, not the GitHub runner, fetches and publishes through its own database connection. A shared durable job with an advisory lock checks source cadence, fetches public arXiv submissions, documented job feeds, provider catalogue pages and DOSM context, uses a RunPod-hosted MiniLM encoder for research matching, validates provenance and atomically publishes eligible snapshots. Stored signals can be rescored without another model call. The older manual GitHub trend path is retained as backup. Page views read stored records; missing rows may use **labelled historical fixture fallback**, not pretend a fresh publication. Optional Gemini embeddings support explicit unfamiliar-skill suggestions. By default the trend matcher switches to Gemini embeddings only after a passing method-equivalence report; a manual operator override is disclosed and stored as a separate series. Optional GLM-5.3 with Gemini failover selects validated Adi answer IDs; matching and plans remain deterministic.



Method: public sources → bounded cleaning → catalogue-ID mapping and paper-mention features → documented heuristic scoring and scenarios → finite/coverage/date validation → dated database snapshots and sourced UI.

3.2 Implementation & Hackathon Deliverables [Maximum 300 words]

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

The Phase 1 submission described a source-grounded journey and proposed evaluation targets; it was not proof of a commercial service. Since the **25 October 2026, 08:00 MYT cutoff**, the team added persisted provenance, bounded daily app-side refresh with recovery and per-source status, clearer empty/stale states, plain-language score tips, one Sources link per card, consent-gated workspaces and programme/funding improvements. Trend signals feed role scores; the Future Skills Radar page shows signal and role-direction cards, while its trend-line charts are deferred to future work. An unchanged-data check is not a new observation. October 2024 research remains historical. Curated roles/courses are not automatically discovered.

The build uses one public web server and a private API, generated contracts, signed sessions, a managed PostgreSQL database and optional RunPod/Gemini integrations. The original GitHub-hosted trend refresher could not reach Replit's private development database; the new authenticated app-side worker is designed to use its own connection. Automatic operation on the public deployment will be confirmed once migrations, configuration and a first scheduled publication are verified. Tests and a local isolated 15-journey synthetic run verify defined source-code paths, not the deployed revision or human outcomes.

The official 48-hour sprint start and end times were not published to the team, so changes are described relative to the 25 October cutoff; the public commit history records exact timestamps.

4. Environmental & SDG Impact Verification

Metric Category	Your Response
Target SDG Metric Achievement (Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers)	Phase 1 baseline (actual Phase 1 PDF, page 10): a problem-context figure, Hays Malaysia 2025 survey, 56% of professionals using AI at work and 26% receiving training (survey methodology not independently verified); these are context, not Adept's pre-intervention KPI. Phase 1 prototype targets: 100% catalogue-ID validity and median intake-to-valid-plan below three minutes across five synthetic demo profiles (not a 20% reduction; no manual planning-time baseline was recorded). Phase 2: an independent catalogue-ID audit and a human timed study are planned for the pilot; the human-trial calculator records 0 of 15 matched cases attempted in each arm. No percentage improvement or measured SDG outcome is claimed.
Verification Evidence Method (How was this data verified?)	An isolated local TestClient run completed 15/15 scripted API journeys on five synthetic profiles, with reported per-persona medians 0.091-0.236 seconds for machine execution; it excluded browser time, human review, network and production data. Source: automated journey record. The human-trial calculator has not yet been run. Catalogue-ID checks require a distinct audit of emitted IDs against authoritative catalogue records.
Socio-Economic / Civic Value (How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?)	Hypothesis: help Malaysian HR/L&D direct training funds toward explained skill gaps, with members retaining choice. Curated target/role SDG mappings offer a reason to choose a learning path, not proof of job placement or SDG 4 indicator gains. The next step is a pilot with a real buyer and independent plan review, after privacy approvals.
Deployment & Commercial Feasibility	Replit-first one-web/one-API plus managed PostgreSQL and shared, not per-view, ingestion. The public prototype has been reachable; matching the deployed revision and a first scheduled publication are the next operational checks. Subscription price, billable worker time, retention, human service hours, legal data rights and procurement cycle are unmeasured; the illustrative calculations in 4.1 are not current profit.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

What the implemented method can prove: Adept's role-matching formula computes a role-fit estimate as `fit = 0.45 × skill_overlap + 0.25 × trend_fit + 0.30 × interest_fit`, where `skill_overlap = Σ importance × min(user_level / required_level, 1) / Σ importance` and a missing interest profile contributes a neutral 0.5. Weights sum to 1, so fit stays in [0, 1]. This is the score before optional SDG weighting: when a learner sets the SDG slider to `w` (0–1), each of the three weights is scaled by `1 − w` and an SDG component is added, giving `fit_sdg = (1 − w) × fit + w × sdg_impact`. The pathway algorithm chooses prerequisite-ordered, catalogue-resolved courses; uncovered gaps remain visible when no course qualifies. This is inspectable calculation, **not** a calibrated job forecast or measured social outcome. A five-synthetic-persona evidence run used a historical 20 October 2026 fixture and an earlier catalogue; its outputs are not current live impact results.

Phase 1 KPI arithmetic (not yet measured): `ID validity % = 100 × resolved recommended course IDs / all recommended course IDs` (undefined if no IDs are recommended). Time target: median over five *human-completed* synthetic-case journeys from intake start to independently reviewed valid plan; report all attempts and failures. A future paired manual comparison may compute `100 × (manual median − Adept median) / manual median`, but **20% was not a Phase 1 target** and the necessary human observations are absent.

Commercial feasibility illustration, not impact proof: hypothesise **RM6 per active seat per month** for a **40-seat, three-month** paid pilot. Revenue if all seats pay throughout = `40 seats × RM6/seat/month × 3 months = RM720`, or **RM240 monthly recognised revenue** (cash collected depends on invoicing). With `v` = monthly delivery cost per customer and `F` = fixed monthly operating cost, contribution = `RM240 − v` and break-even paid customers = `ceil(F / (RM240 − v))` only if `v < RM240`. Neither `v` nor `F` is measured;

neither margin nor break-even customer count is presently known. HRD Corp training levies do **not** automatically pay for a software subscription.

5. Market Viability & Future Scalability Plan [Maximum 250 words]

Adept's initial **hypothesis** is an add-on for Malaysian employer HR/L&D teams planning training for AI-exposed operations staff, not an HR/payroll system or employment predictor. HRD Corp confirms a 1% training levy for covered employers with at least ten Malaysian employees; this proves a training budget exists, **not** that Adept subscriptions are claimable or employers will buy. BrioHR lists training administration at RM3 per month with a RM200 minimum and a talent bundle at RM7; Adept must prove its gap-first planning is worth an additional fee.

An eight-to-twelve-week, 25–50-seat pilot would compare manual course selection with independently checked plans, decision time, relevance, consent and support hours. RM6 per active learner per month is an **unvalidated test price**: 40 paid seats for three months would bill RM720, without automatically covering delivery. Renewal requires buyer-confirmed value. No interview, purchase or revenue is claimed.

The existing system shares catalogue and ingestion work, caches dated snapshots, uses deterministic scoring on reads and can process asynchronous refresh jobs against Replit-managed PostgreSQL. Scale-up requires measured concurrency, database/storage bills, worker seconds, restore drills, licence permissions, privacy agreements and human review capacity. Sponsor credits and unpaid team labour are temporary. Only measured bottlenecks should trigger additional workers or a service split. Employers retain training and personnel decisions. Market evidence (HRD Corp's RM2.62bn 2025 training approvals; 4.2 million highly AI-exposed workers) is cited in the accompanying PRD.

6. Final Submission Checklist

☐ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.

☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.

☐ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.

☐ The final pitch deck has been exported and saved as a flattened PDF file.

☐ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

To be completed by team leader Team record as of 25 October 2026: no pre-existing work is included. Adept's implementation was created for the hackathon; the separate KerjaCerdas project was consulted only for general security ideas, and no code was imported.

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

Name	Type (code/ dataset/ model)	License
Next.js, React, Tailwind CSS, next-intl, Recharts, Three.js, FastAPI, Pydantic, Uvicorn, HTTPX	code	MIT, except Uvicorn and HTTPX BSD-3-Clause
PostgreSQL, pgvector, psycopg, PyMuPDF	code	PostgreSQL License; psycopg LGPL-3.0; PyMuPDF AGPL-3.0 or Artifex commercial licence (reviewed before any commercial distribution)
arXiv cs.AI metadata, US BLS projections, O*NET 31.0, UN SDG definitions, OpenDOSM Labour Force Survey, MyCOL, public job feeds, provider course facts	dataset	arXiv metadata CC0 1.0; BLS US public domain; O*NET and OpenDOSM CC BY 4.0 with attribution; UN and TalentCorp terms of use; job ads and course facts under each provider's terms
Sentence Transformers all-MiniLM-L6-v2; optional GLM-5.3 (Z.ai, via RunPod); optional Gemini models and gemini-embedding-001	model	MiniLM Apache-2.0; GLM-5.3 under Z.ai and RunPod terms; Gemini under the Gemini API Terms of Service
Synthetic personas, curated skills, roles, course links and SDG alignments	dataset	Team-created; underlying provider and taxonomy rights retained; not an official UN endorsement

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

To be completed by team leader

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

Signature: _____

Date: _____